b) $f(S \cap T) \subseteq f(S) \cap f(T)$

Solution: $S \cap T = \{x / x \in S \wedge x \in T\}$ (By Defn. 2 from Sec-2.2)

Let $a \in f(S \cap T)$. $\therefore \exists y \in S \cap T$ s.t. $f(y) = a$.

By Defn. 2 from Sec-2.2, $y \in S \wedge y \in T$. By Defn. of AND, $y \in S$ and $y \in T$ are both True. $\therefore a \in f(S)$ and $a \in f(T)$. By Defn. of AND, $a \in f(S) \wedge f(T)$.

By Defn. 2 from Sec-2.2, $a \in f(S) \cap f(T)$. By Defn. 3 from Sec-2.1,

$f(S \cap T) \subseteq f(S) \cap f(T)$ □

43) a) Give an example to show that the inclusion in part (b) in Exercise 42 may be proper.

Solution: $A = \{1, 2, 3, 4\}$ $B = \{a, b, c, d\}$.

$S = \{1, 2, 3\}$ $T = \{2, 4\}$

$f(1) = b$ $f(2) = d$ $f(3) = a$ $f(4) = a$

$S \cap T = \{2\}$ $f(S \cap T) = \{d\}$

$f(S) = \{b, d, a\}$ $f(T) = \{d, a\}$

$f(S) \cap f(T) = \{d, a\}$. $\therefore f(S \cap T) \subset f(S) \cap f(T)$

$\therefore \{d\} \subset \{d, a\}$ □

b) Show that if f is one-to-one, the inclusion in part (b) in Exercise 42 is an equality.

Solution: By Defn. 5, if $x, y \in A$ and $f(x) = f(y)$, then $x = y$.

Let $a \in b$ and $a \in f(S) \cap f(T)$. By Defn. 2 from Sec-2.2, $a \in f(S) \wedge f(T)$.

By Defn. of AND, $a \in f(S)$ and $a \in f(T)$. $\therefore \exists n \in S$ and $m \in T$ s.t.

$f(n) = a$ and $f(m) = a$. By Defn. 5, $m = n$.

$\therefore$ By Defn. of AND, $n \in S \wedge n \in T$. By Defn. 2, $n \in S \cap T$.

$\therefore f(n) \in f(S \cap T) \Rightarrow a \in f(S \cap T)$.

By Defn. 3 from Sec-2.1, $f(S) \cap f(T) \subseteq f(S \cap T)$.

By Prob 42(b), we have $f(S \cap T) = f(S) \cap f(T)$ □